

Ben Morris

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Pittsburgh, PA | Able to relocate

A creative engineering student with hands-on experience in robotics and software development, passionate about integrating digital systems and physical hardware to build imaginative, technology-driven solutions. Skilled in developing cutting-edge tools for entertainment and interactive technology.

Education

Master of Entertainment Technology Aug 2025 - May 2027

Carnegie Mellon University | Entertainment Technology Center | Pittsburgh, PA | 3.80 GPA

B.S. in Engineering with a Concentration in Computing

Aug 2020 - May 2024

Olin College of Engineering | Needham, MA | 3.80 GPA

Skills

Software: Python, Java, C#, C, C++, Arduino, MATLAB, OCaml, Haskell, Swift, Javascript, SQL, SystemVerilog

Systems: Git/GitHub, Perforce, ROS2, Unity, Ubuntu, Computer Vision, LiDAR, Slack, Jira, Notion, Trello

Design: Figma, Lucidchart, Adobe Illustrator, Photoshop, Premiere Pro

Skills: Project Management, Scrum/Agile, Team Leadership, Software/Hardware Integration, Experience Design

Languages: Native fluency in English, Conversational in French

Work Experience

Robotics Software Engineer | Tutor Intelligence | Watertown, MA

Dec 2024 - Jul 2025

- Engineered new core features to improve robustness in automatic palletizing solutions
- Collaborated with a 25 person cross-disciplinary team to set technical direction, establish milestones, and integrate software/hardware components
- Spearheaded effort to mitigate mechanical failures in sensors, implementing automatic issue resolution and early warning signs for failure, reducing mechanical sensor failures by more than 80%
- Revamped manipulation technique, achieving a 1.4x average improvement in grip strength across SKUs

Academic Experience

Lead Drone Research Software Developer | RoboLab | Needham, MA

May 2022 - Dec 2022

- Researched and implemented localization and efficient path planning algorithms in a tight environment
- Mentored a team of 7 students controlling a drone using ArduPilot in Python/C with Marvelmind localization
- Created a custom real-time path planning algorithm for a swarm of drones which avoided collisions in a tight environment, and achieving top speed 70% of the time with a 3 cm error margin

Software Researcher | OCCaM Lab | Needham, MA

May 2021 - Dec 2021

- Developed an iOS AR app to assist the blind community in independently navigating unfamiliar buildings, with replicable map creation & path finding functionality for scalability
- Integrated LiDAR, April Tags, & V/IO in Swift/Python, solving SLAM problems to create maps of buildings
- Built a real-time raycasting system to find navigational shortcuts within generated maps
- Advanced project from proof-of-concept to functional prototype, achieving a 75% improvement in accuracy

Relevant Projects

President, Franklin W. Olin Players: Led 40+ students across 6 plays/musicals as director, stage manager, & actor

Software Engineer, Neato Tag: Developed a ROS2-based multi-robot swarm to play tag by integrating LiDAR & CV for real-time localization and target tracking, demonstrating reliable chase and evade behavior

Software Engineer, Altitude: Designed & developed an immersive Unity skydiving game using a custom full-body motion controller to simulate freefall control